openwifi 移植指南

长春顺为达科技有限公司

目　录

前提：安装虚拟机 Ubuntu 或 Ubuntu 系统、Vivado2022.2、Vitis2022.2

1 获取源码与镜像文件

- 硬件设计仓库(FPGA 源码)

  git clone --recursive https://github.com/open-sdr/openwifi-hw

- 软件设计仓库(嵌入式 Linux 源码)

  git clone --recursive https://github.com/open-sdr/openwifi

- 下载 openwifi.img

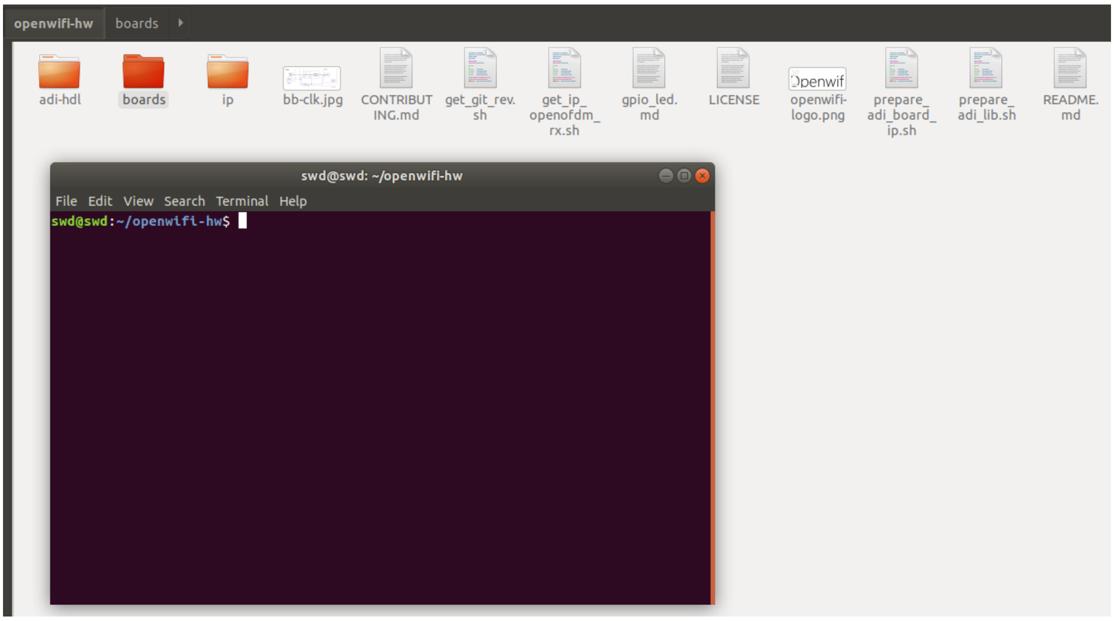
Download img and Quick start

- Download openwifi img, unzip and burn it into a SD card (>=16GB). After this operation, the SD card should have two partitions: BOOT and rootfs. To flash the SD card, SD card tool software (such as Startup Disk Creator in Ubuntu) or dd command can be used:

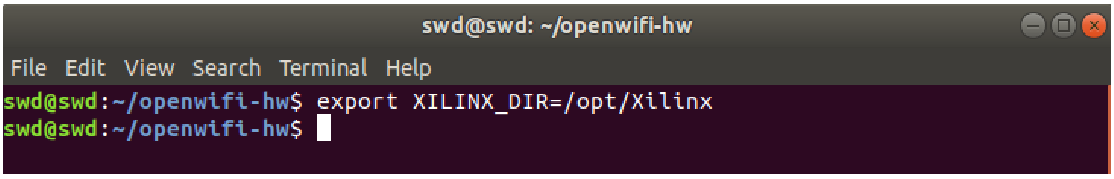
2 构建 FPGA 工程

选择 openwifi 已支持的最接近的参考平台，构建 FPGA 工程

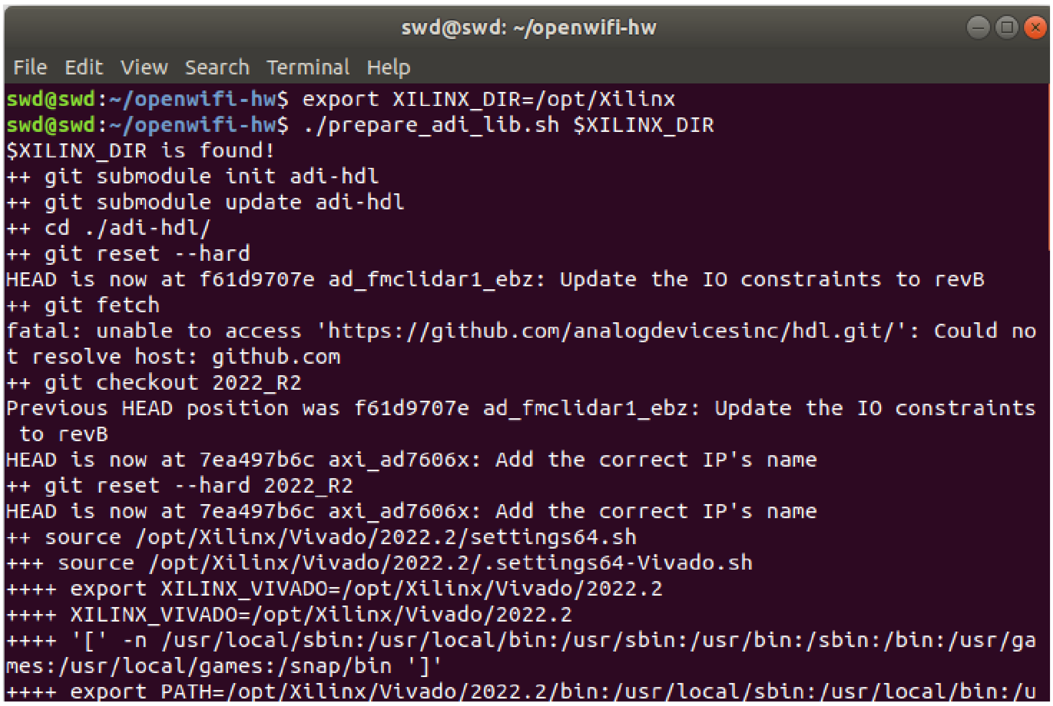
笔者选择参考 zc702_fmcs2 平台，进入 openwifi_hw 文件夹打开终端。



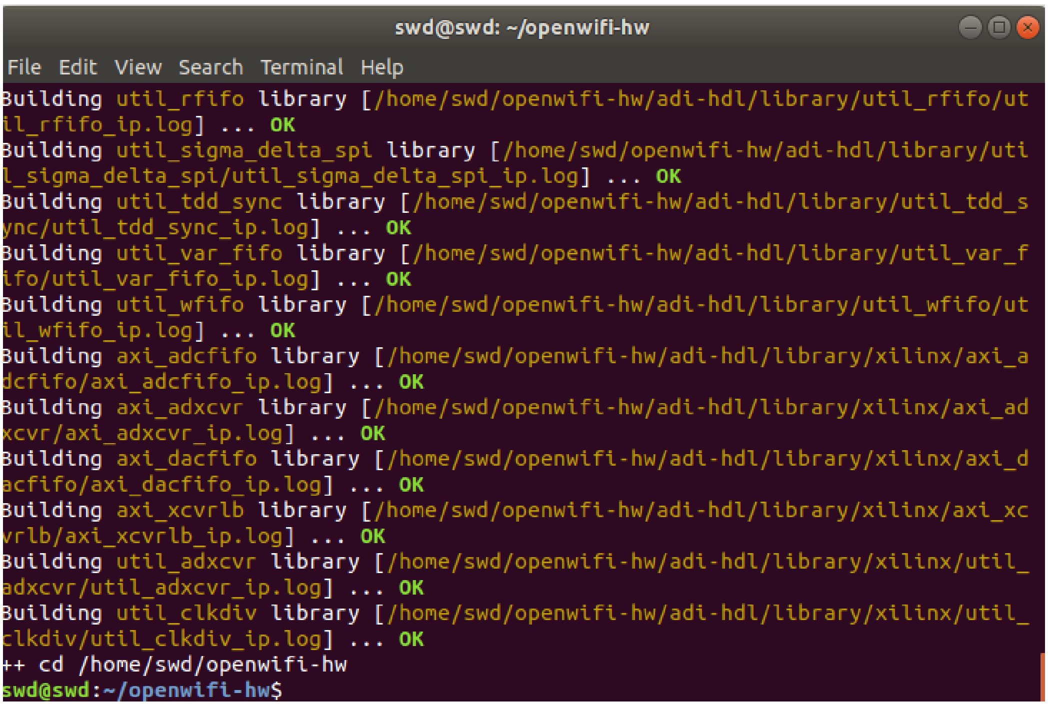
- 输入 Xilinx 软件位置，笔者的安装在/opt/Xilinx 文件夹
- export XILINX_DIR=/opt/Xilinx



- 构建 HDL 库
- ./prepare_adi_lib.sh $XILINX_DIR



- 时间较长，每个库构建成功后会有提示，如下图。



- 选择参考平台
- export_BOARD_NAME=zc702_fmcs2

```
swd@swd:~/openwifi-hw$ export BOARD_NAME=zc702_fmcs2
swd@swd:~/openwifi-hw$
```

- 构建 IP 库
- ./prepare_adi_board_ip.sh $XILINX_DIR $BOARD_NAME

```
swd@swd:~/openwifi-hw$ ./prepare_adi_board_ip.sh $XILINX_DIR $BOARD_NAME
$XILINX_DIR is found!
./adi-hdl/projects/fmcomms2/zc702/
+ source /opt/Xilinx/Vivado/2022.2/settings64.sh
++ source /opt/Xilinx/Vivado/2022.2/.settings64-Vivado.sh
+++ export XILINX_VIVADO=/opt/Xilinx/Vivado/2022.2
+++ XILINX_VIVADO=/opt/Xilinx/Vivado/2022.2
+++ '[' -n /usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin ']'
+++ export PATH=/opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
+++ PATH=/opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
++ source /opt/Xilinx/Vitis/2022.2/.settings64-Vitis.sh
+++ export XILINX_VITIS=/opt/Xilinx/Vitis/2022.2
+++ XILINX_VITIS=/opt/Xilinx/Vitis/2022.2
+++ '[' -n /opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin ']'
+++ export PATH=/opt/Xilinx/Vitis/2022.2/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/arm/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/linux_t
oolchain/lin64_le/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-linux-gnueabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-none-eabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/
aarch64/lin/aarch64-linux/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-none/bin:/opt/Xilinx/Vitis/2022.2/gnu/armr5/lin/gcc-arm-none-eabi/bin:/opt/Xilinx/Vitis/2022.2/tps/lnx
64/cmake-3.3.2/bin:/opt/Xilinx/Vitis/2022.2/aietools/bin:/opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap
/bin
+++ PATH=/opt/Xilinx/Vitis/2022.2/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/arm/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/linux_toolchai
n/lin64_le/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-linux-gnueabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-none-eabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64
/lin/aarch64-linux/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-none/bin:/opt/Xilinx/Vitis/2022.2/gnu/armr5/lin/gcc-arm-none-eabi/bin:/opt/Xilinx/Vitis/2022.2/tps/lnx64/cmak
e-3.3.2/bin:/opt/Xilinx/Vitis/2022.2/aietools/bin:/opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
++ source /opt/Xilinx/Vitis_HLS/2022.2/.settings64-Vitis_HLS.sh
+++ export XILINX_HLS=/opt/Xilinx/Vitis_HLS/2022.2
+++ XILINX_HLS=/opt/Xilinx/Vitis_HLS/2022.2
+++ '[' -n /opt/Xilinx/Vitis/2022.2/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/arm/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/linux_toolch
ain/lin64_le/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-linux-gnueabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-none-eabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch
64/lin/aarch64-linux/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-none/bin:/opt/Xilinx/Vitis/2022.2/gnu/armr5/lin/gcc-arm-none-eabi/bin:/opt/Xilinx/Vitis/2022.2/tps/lnx64/cm
ake-3.3.2/bin:/opt/Xilinx/Vitis/2022.2/aietools/bin:/opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/games:/usr/local/games:/snap/bin
']'
+++ export PATH=/opt/Xilinx/Vitis_HLS/2022.2/bin:/opt/Xilinx/Vitis/2022.2/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/arm/lin/bin:/opt/Xilinx/Vi
tis/2022.2/gnu/microblaze/linux_toolchain/lin64_le/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-linux-gnueabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-none-eabi/
bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-linux/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-none/bin:/opt/Xilinx/Vitis/2022.2/gnu/armr5/lin/gcc-arm-none-eabi/bin
:/opt/Xilinx/Vitis/2022.2/tps/lnx64/cmake-3.3.2/bin:/opt/Xilinx/Vitis/2022.2/aietools/bin:/opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:
/usr/games:/usr/local/games:/snap/bin
+++ PATH=/opt/Xilinx/Vitis_HLS/2022.2/bin:/opt/Xilinx/Vitis/2022.2/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/arm/lin/bin:/opt/Xilinx/Vitis/202
2.2/gnu/microblaze/linux_toolchain/lin64_le/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-linux-gnueabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-none-eabi/bin:/op
t/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-linux/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-none/bin:/opt/Xilinx/Vitis/2022.2/gnu/armr5/lin/gcc-arm-none-eabi/bin:/opt/X
ilinx/Vitis/2022.2/tps/lnx64/cmake-3.3.2/bin:/opt/Xilinx/Vitis/2022.2/aietools/bin:/opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/ga
mes:/usr/local/games:/snap/bin
+ cd ./adi-hdl/projects/fmcomms2/zc702/
+ make
Building fmcomms2_zc702 project [/home/swd/openwifi-hw/adi-hdl/projects/fmcomms2/zc702/fmcomms2_zc702_vivado.log] ...
```

- 当出现 Building xxx project……时，ctrl+c 直接取消即可。

```
Building fmcomms2_zc702 project [/home/swd/openwifi-hw/adi-hdl/projects/fmcomms2/zc702/fmcomms2_zc702_vivado.log] ...^C
../../scripts/project-xilinx.mk:100: recipe for target 'fmcomms2_zc702.sdk/system_top.xsa' failed
make: *** [fmcomms2_zc702.sdk/system_top.xsa] Interrupt

swd@swd:~/openwifi-hw$
swd@swd:~/openwifi-hw$
```

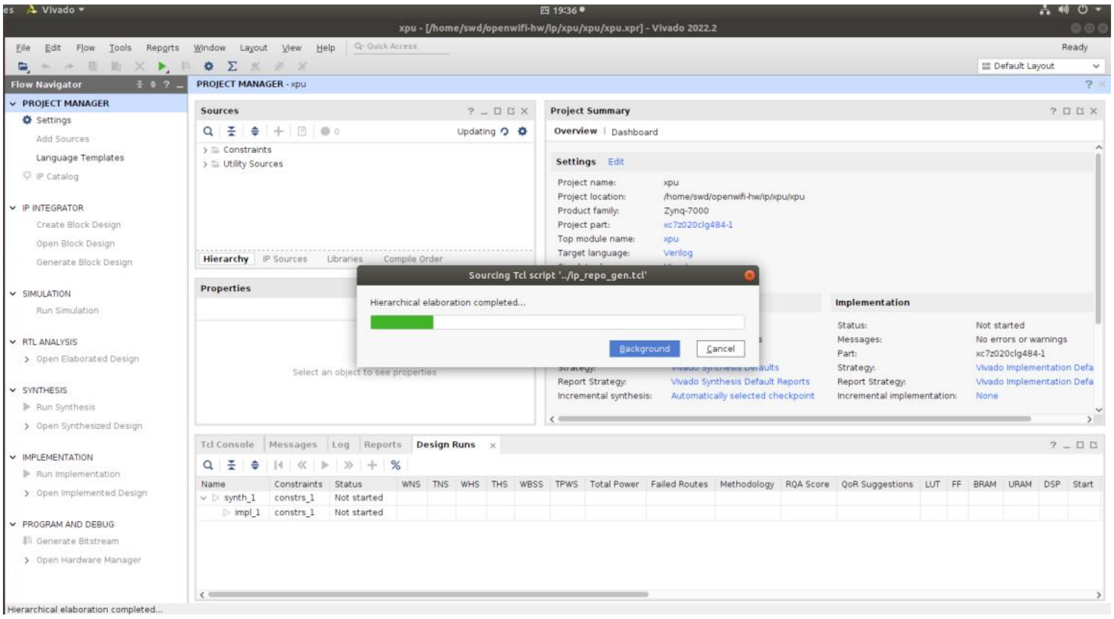
- 更新 ofdmrx IP
- ./get_ip_openofdm_rx.sh

```
swd@swd:~/openwifi-hw$ ./get_ip_openofdm_rx.sh
+ cd ip/
+ git submodule init openofdm_rx
+ git submodule update openofdm_rx
+ cd /home/swd/openwifi-hw
swd@swd:~/openwifi-hw$
```

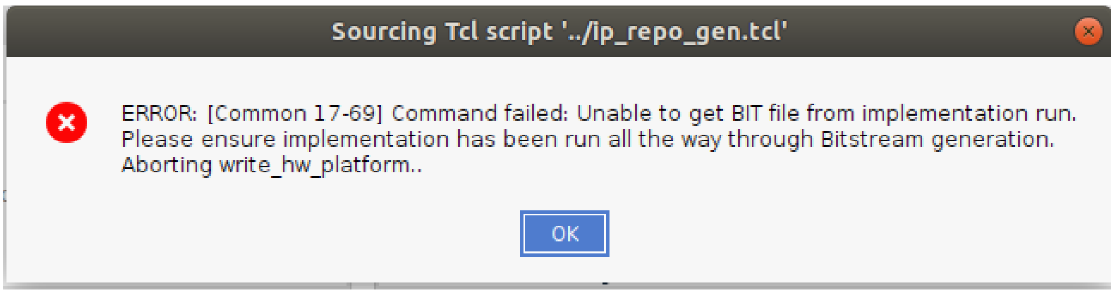
- cd boards/$BOARD_NAME/
- ../create_ip_repo.sh $XILINX_DIR

```
swd@swd:~/openwifi-hw$ cd boards/$BOARD_NAME/
swd@swd:~/openwifi-hw/boards/zc702_fmcs2$ ../create_ip_repo.sh $XILINX_DIR
usage:
create_ip_repo.sh $XILINX_DIR
or
create_ip_repo.sh $XILINX_DIR $IP1_NAME $DEF1 $DEF2 ... $IP2_NAME $DEF1 ...
 -IP_NAME: only xpu/tx_intf/rx_intf/openofdm_tx/openofdm_rx/side_ch are allowed
 -   DEFx: will be "`define IP_NAME_DEFx" in ip_name_pre_def.v for $IP_NAME

zc702_fmcs2
Expect env file /opt/Xilinx/Vitis/2022.2/settings64.sh
/opt/Xilinx/Vitis/2022.2/settings64.sh is found!
+ vivado -source ../ip_repo_gen.tcl

****** Vivado v2022.2 (64-bit)
  **** SW Build 3671981 on Fri Oct 14 04:59:54 MDT 2022
  **** IP Build 3669848 on Fri Oct 14 08:30:02 MDT 2022
    ** Copyright 1986-2022 Xilinx, Inc. All Rights Reserved.

start_gui
```

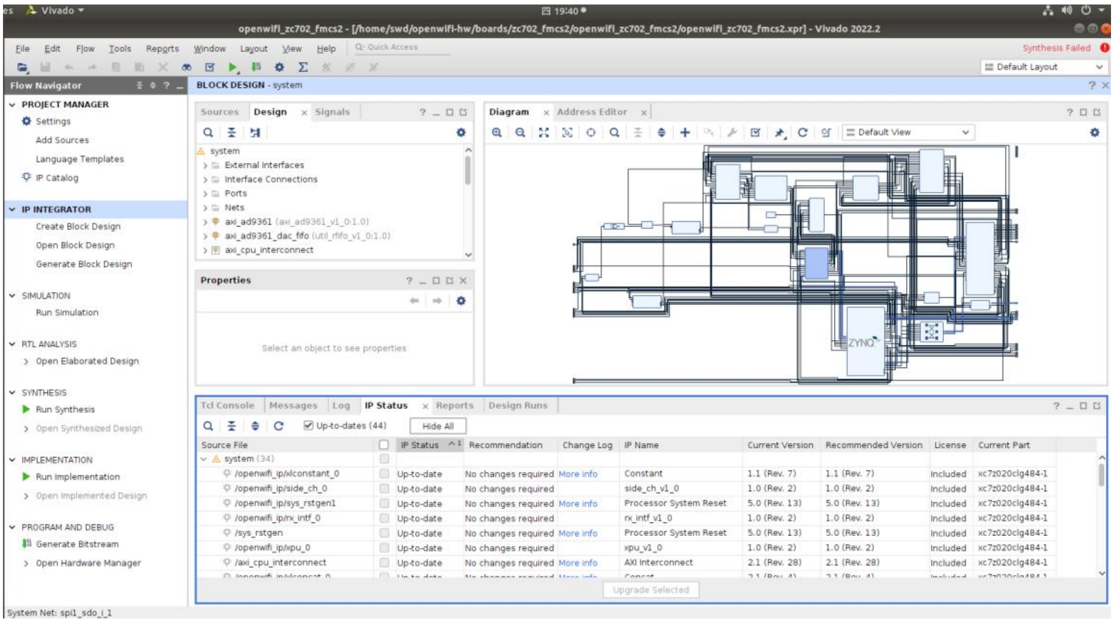
➢ 自动打开 Vivado 构建工程，时间较长，不要手动干预。



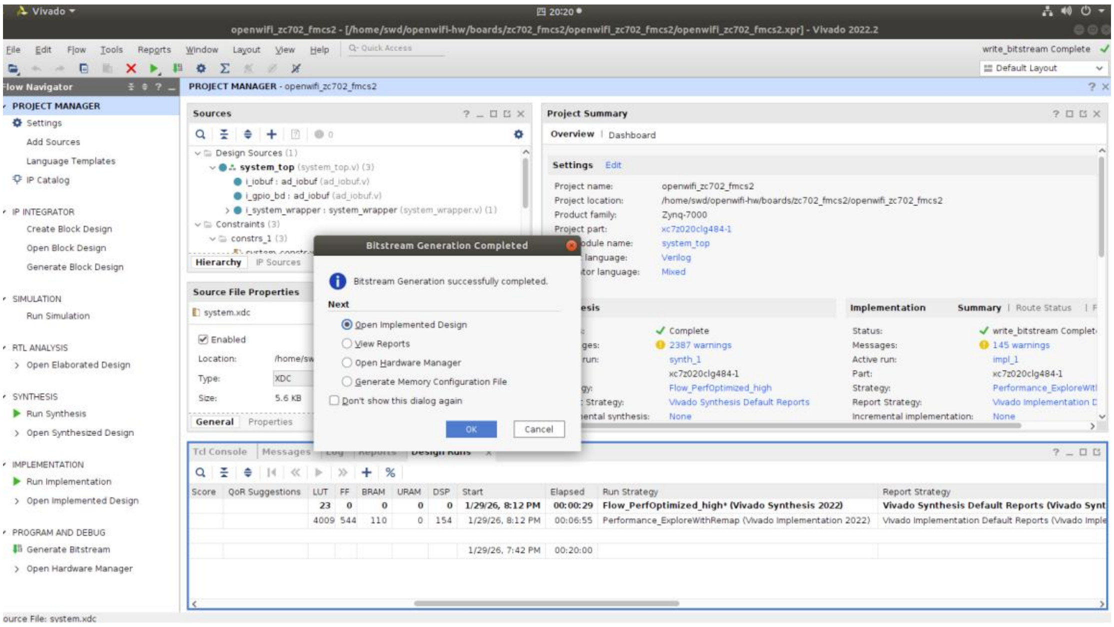
➢ 出现如下提示，关闭即可。



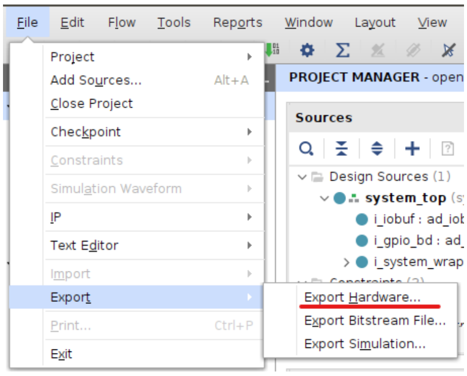
➢ 在参考工程的基础上，更改硬件设计，比如 PS 外设配置更改、引脚分配更改、删除多余 IP 等，适配要移植的板卡。

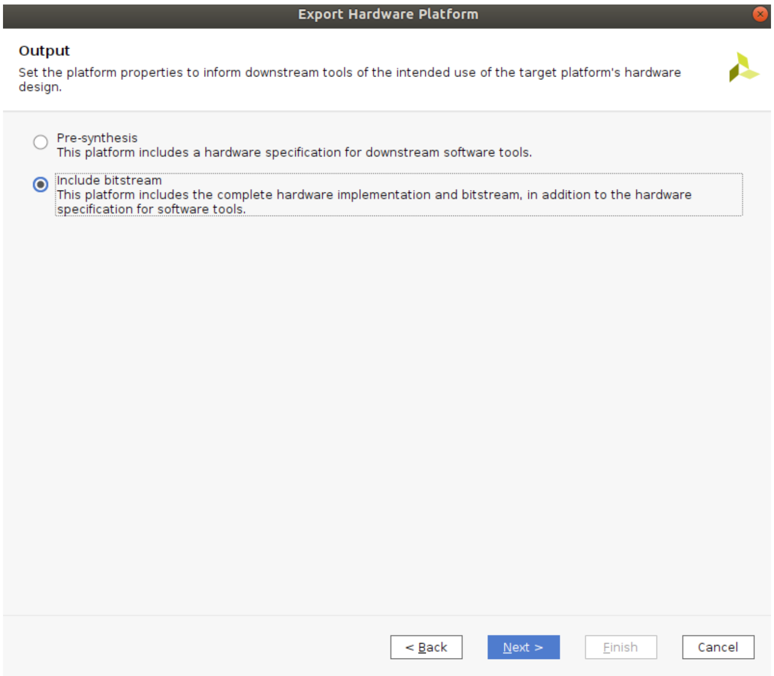


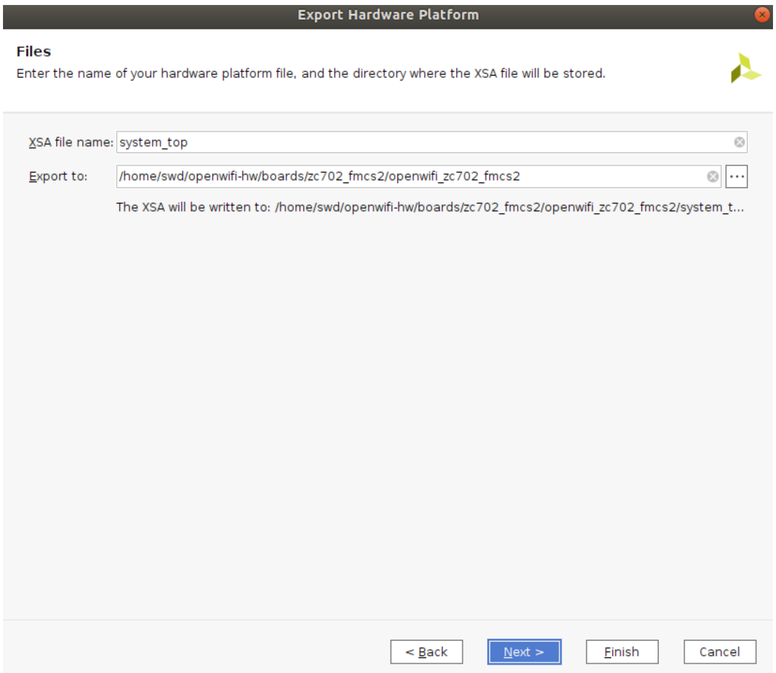
- 更改完成后，编译工程，生成 bitstream，时间较长，笔者的工程编译了 7 分钟左右。

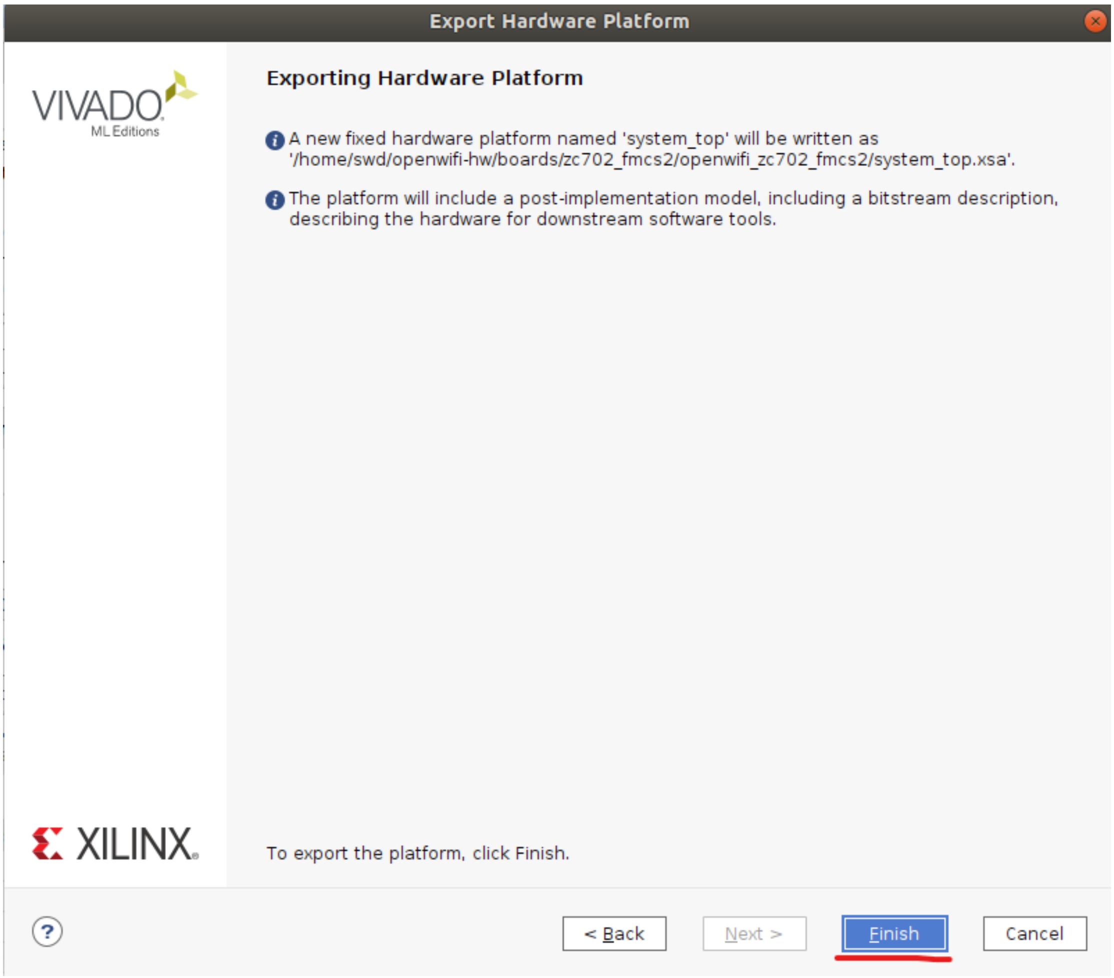


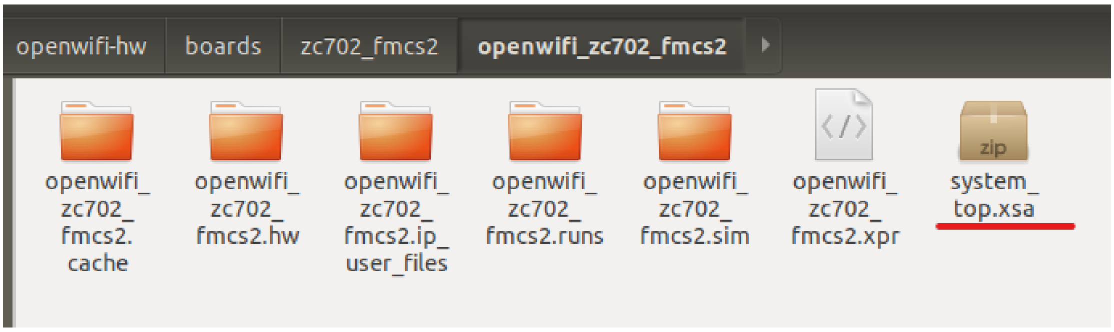
➢ 导出硬件信息.xsa 文件到工程根目录。









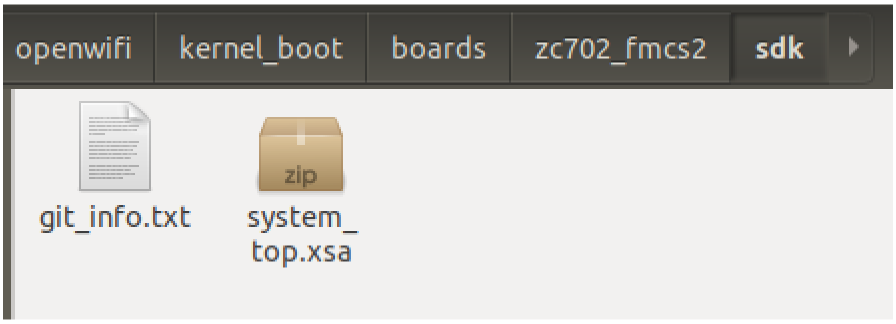


3 生成 bin 文件与设备树

- cd ..
- export OPENWIFI_HW_IMG_DIR=/home/swd/openwifi/kernel_boot
- ./sdk_update.sh &BOARD_NAME & OPENWIFI_HW_IMG_DIR

```
swd@swd:~/openwifi-hw/boards/zc702_fmcs2$ cd ..
swd@swd:~/openwifi-hw/boards$ export OPENWIFI_HW_IMG_DIR=/home/swd/openwifi/kernel_boot
swd@swd:~/openwifi-hw/boards$ ./sdk_update.sh $BOARD_NAME $OPENWIFI_HW_IMG_DIR
+ BOARD_NAME=zc702_fmcs2
+ OPENWIFI_HW_IMG_DIR=/home/swd/openwifi/kernel_boot
+ echo zc702_fmcs2
zc702_fmcs2
+ echo /home/swd/openwifi/kernel_boot
/home/swd/openwifi/kernel_boot
+ TARGET_SDK_DIR=/home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/
+ test -d /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/
+ echo 'Create /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/'
Create /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/
+ mkdir /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/ -p
+ rm -rf '/home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk//*'
+ cp zc702_fmcs2/openwifi_zc702_fmcs2/system_top.xsa /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk// -rf
+ '[' -f zc702_fmcs2/openwifi_zc702_fmcs2/openwifi_zc702_fmcs2.runs/impl_1/system_top.ltx ']'
+ echo 'No debug probe file found.'
No debug probe file found.
+ echo openwifi-hw-git-branch
+ git branch
+ echo ' '
+ echo openwifi-hw-git-commit
+ git log -3
+ echo ' '
+ echo openofdm_rx-git-branch
+ git --git-dir ../ip/openofdm_rx/.git branch
+ echo ' '
+ echo openofdm_rx-git-commit
+ git --git-dir ../ip/openofdm_rx/.git log -3
+ echo ' '
swd@swd:~/openwifi-hw/boards$
```

➢ 查看 openwifi 文件夹，看到.xsa 文件已经同步更新。



➢ 生成 BOOT.bin 和 system_top.bit.bin 文件。

➢ cd ../../

➢ cd openwifi/user_space

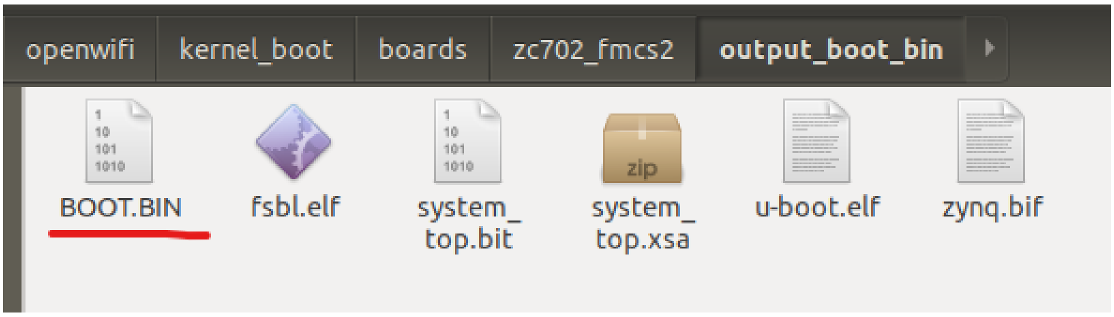
➢ ./boot_bin_gen.sh $XILINX_DIR $BOARD_NAME …

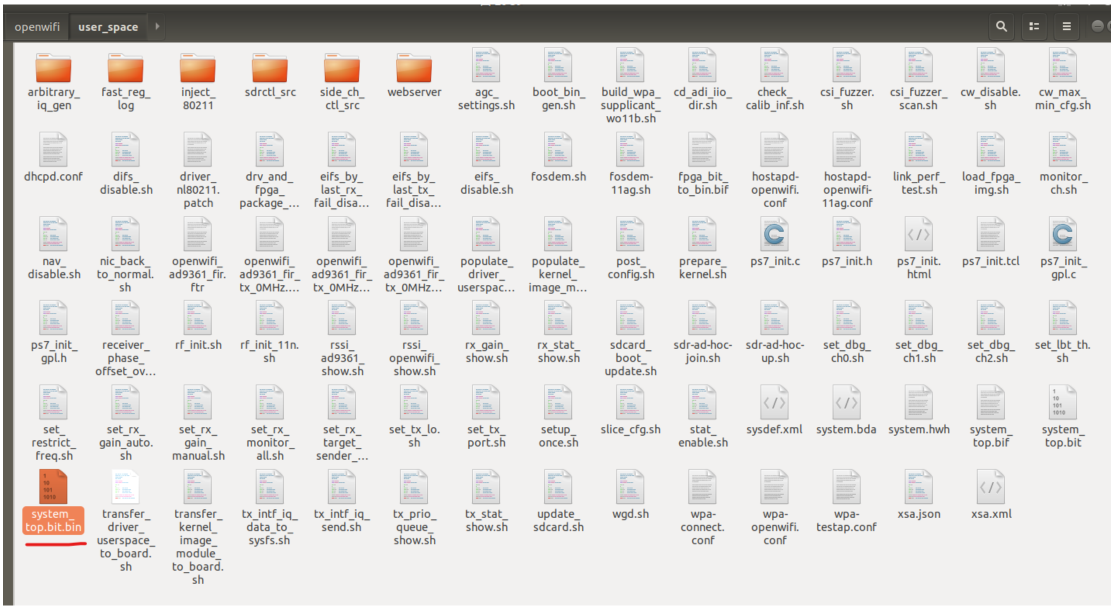
$OPENWIFI_HW_IMG_DIR/boards/$BOARD_NAME/sdk/system_top.xsa

```
swd@swd:~/openwifi-hw/boards$ cd ../../
swd@swd:~$ cd openwifi/user_space
swd@swd:~/openwifi/user_space$ ./boot_bin_gen.sh $XILINX_DIR $BOARD_NAME $OPENWIFI_HW_IMG_DIR/boards/$BOARD_NAME/sdk/system_top.xsa
OPENWIFI_DIR /home/swd/openwifi/user_space/../
XSA_FILE /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/system_top.xsa
/home/swd/openwifi/user_space/../ is found!
Expect env file /opt/Xilinx/Vitis/2022.2/settings64.sh
/opt/Xilinx/Vitis/2022.2/settings64.sh is found!
/home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/system_top.xsa is found!
+ source /opt/Xilinx/Vitis/2022.2/settings64.sh
++ source /opt/Xilinx/Vivado/2022.2/.settings64-Vivado.sh
+++ export XILINX_VIVADO=/opt/Xilinx/Vivado/2022.2
+++ XILINX_VIVADO=/opt/Xilinx/Vivado/2022.2
+++ '[' -n /opt/Xilinx/Vitis_HLS/2022.2/bin:/opt/Xilinx/Vitis/2022.2/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/arm/lin/bin:/opt/Xilinx/Vitis/2
022.2/gnu/microblaze/linux_toolchain/lin64_le/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-linux-gnueabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-none-eabi/bin:/
opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-linux/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-none/bin:/opt/Xilinx/Vitis/2022.2/gnu/armr5/lin/gcc-arm-none-eabi/bin:/opt
/Xilinx/Vitis/2022.2/tps/lnx64/cmake-3.3.2/bin:/opt/Xilinx/Vitis/2022.2/aietools/bin:/opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin:/usr/
games:/usr/local/games:/snap/bin ']'
+++ export PATH=/opt/Xilinx/Vivado/2022.2/bin:/opt/Xilinx/Vitis_HLS/2022.2/bin:/opt/Xilinx/Vitis/2022.2/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/lin/bin:/opt/Xilinx/Vitis/2022.2/
gnu/arm/lin/bin:/opt/Xilinx/Vitis/2022.2/gnu/microblaze/linux_toolchain/lin64_le/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch32/lin/gcc-arm-linux-gnueabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/
aarch32/lin/gcc-arm-none-eabi/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-linux/bin:/opt/Xilinx/Vitis/2022.2/gnu/aarch64/lin/aarch64-none/bin:/opt/Xilinx/Vitis/2022.2/gnu/a
rmr5/lin/gcc-arm-none-eabi/bin:/opt/Xilinx/Vitis/2022.2/tps/lnx64/cmake-3.3.2/bin:/opt/Xilinx/Vitis/2022.2/aietools/bin:/opt/Xilinx/Vivado/2022.2/bin:/usr/local/sbin:/usr/local/bin:
```

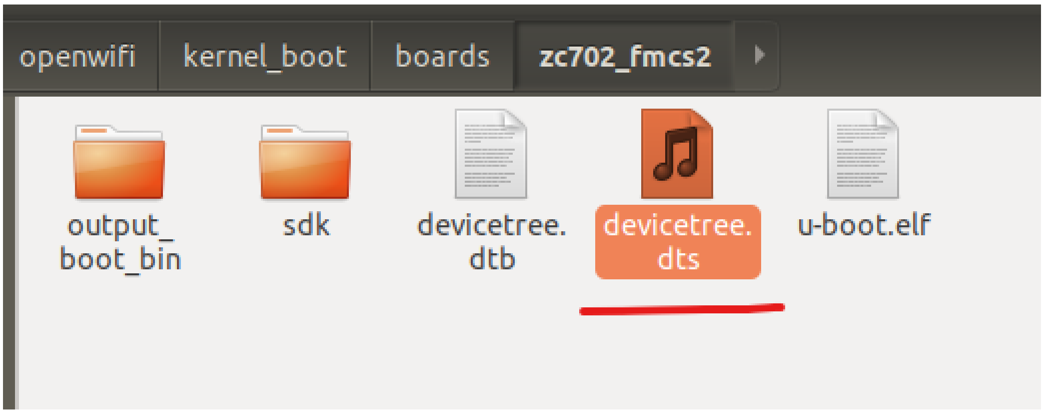
```
+ '[' 0 -ne 0 ']'
+ ARCH=zynq
+ ARCH_BIT=32
+ rm -rf build_boot_bin
+ rm -rf boards/zc702_fmcs2/output_boot_bin
+ mv output_boot_bin boards/zc702_fmcs2/
+ cd /home/swd/openwifi/user_space
++ basename /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/system_top.xsa
+ XSA_FILE_BASENAME=system_top.xsa
++ basename /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/system_top.xsa .xsa
+ XSA_FILE_BASENAME_WO_EXT=system_top
+ BIT_FILE_BASENAME=system_top.bit
+ echo all:
+ echo '{'
+ echo 'system_top.bit /* Bitstream file name */'
+ echo '}'
+ unzip -o /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/system_top.xsa
Archive:  /home/swd/openwifi/kernel_boot/boards/zc702_fmcs2/sdk/system_top.xsa
E47w6VK0RQkcHWGA8SfkPL/Ib3b34EMgvxuoGt1X2m7Tk=
  inflating: sysdef.xml
  inflating: ps7_init.h
  inflating: ps7_init.c
  inflating: xsa.json
  inflating: system.hwh
  inflating: ps7_init.html
  inflating: xsa.xml
  inflating: system.bda
  inflating: ps7_init_gpl.c
  inflating: ps7_init.tcl
  inflating: system_top.bit
  inflating: ps7_init_gpl.h
+ rm -rf ./system_top.bit.bin
+ bootgen -image fpga_bit_to_bin.bif -arch zynq -process_bitstream bin -w


****** Xilinx Bootgen v2022.2.0
  **** Build date : Oct 13 2022-12:22:43
    ** Copyright 1986-2022 Xilinx, Inc. All Rights Reserved.


[INFO]   : Bootimage generated successfully

+ ls ./system_top.bit.bin -al
-rw-rw-r-- 1 swd swd 4045568 1月  29 20:37 ./system_top.bit.bin
swd@swd:~/openwifi/user_space$
```

➢ 查看生成的文件。

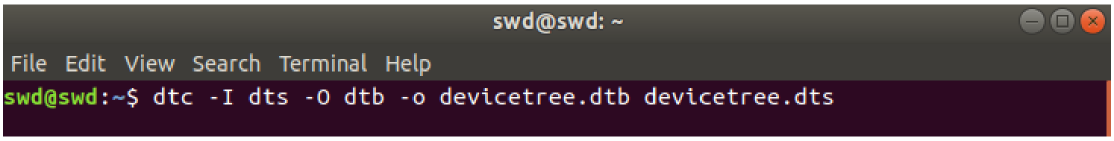




- 找到参考工程的设备树文件，修改设备树内容，适配要移植的板卡。



- 生成 dtb 文件，没有 DTC 工具先安装工具 sudo apt-get install device-tree-compiler。
- dtc -I dts -O dtb -o devicetree.dtb devicetree.dts

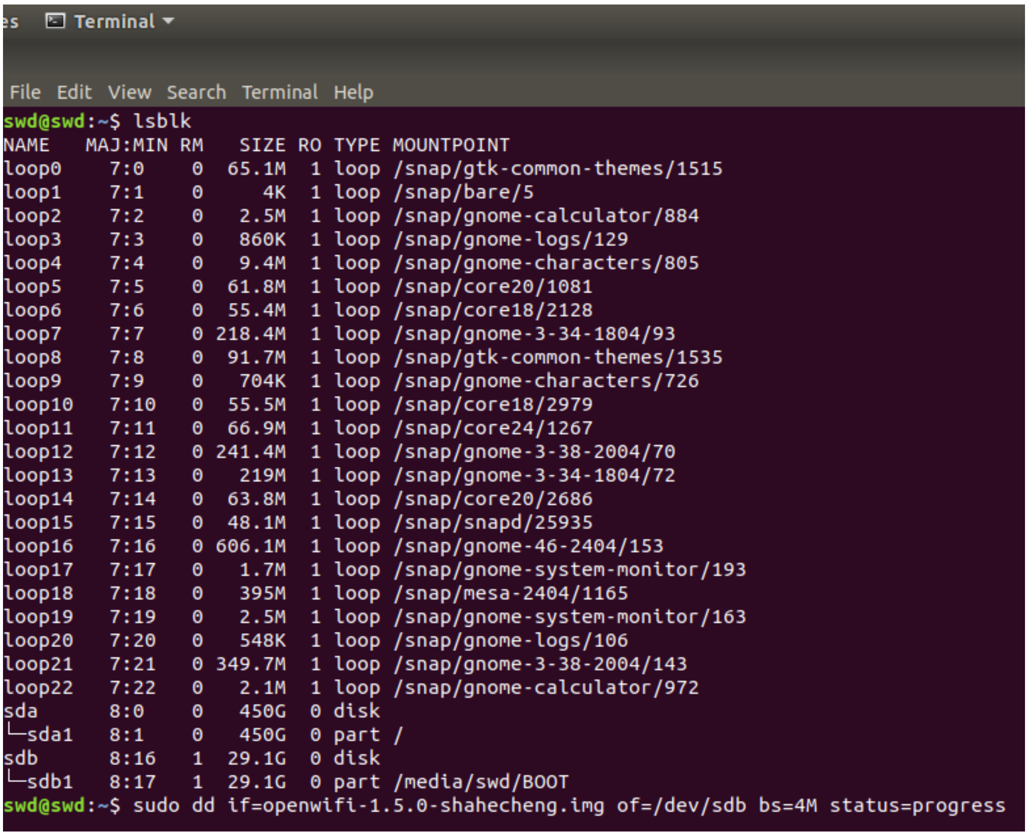


4 SD 卡镜像烧录与设置

- 准备好的 SD 卡镜像文件，烧录到 SD 卡中。
- lsblk 找到 SD 卡设备，笔者的为 sdb。

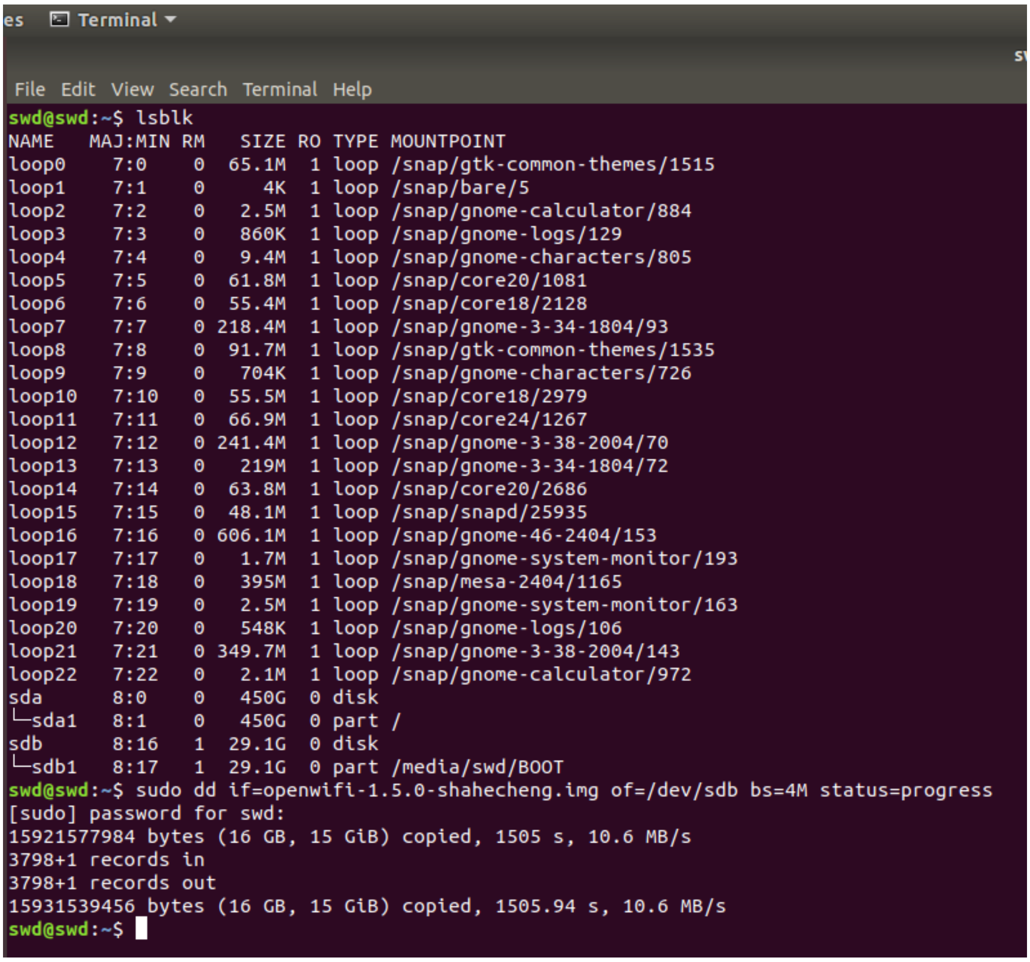
```
swd@swd:~$ lsblk
NAME    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
loop0     7:0    0  65.1M  1 loop /snap/gtk-common-themes/1515
loop1     7:1    0    4K  1 loop /snap/bare/5
loop2     7:2    0   2.5M  1 loop /snap/gnome-calculator/884
loop3     7:3    0   860K  1 loop /snap/gnome-logs/129
loop4     7:4    0   9.4M  1 loop /snap/gnome-characters/805
loop5     7:5    0  61.8M  1 loop /snap/core20/1081
loop6     7:6    0  55.4M  1 loop /snap/core18/2128
loop7     7:7    0 218.4M  1 loop /snap/gnome-3-34-1804/93
loop8     7:8    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop9     7:9    0   704K  1 loop /snap/gnome-characters/726
loop10    7:10   0  55.5M  1 loop /snap/core18/2979
loop11    7:11   0  66.9M  1 loop /snap/core24/1267
loop12    7:12   0 241.4M  1 loop /snap/gnome-3-38-2004/70
loop13    7:13   0   219M  1 loop /snap/gnome-3-34-1804/72
loop14    7:14   0  63.8M  1 loop /snap/core20/2686
loop15    7:15   0  48.1M  1 loop /snap/snapd/25935
loop16    7:16   0 606.1M  1 loop /snap/gnome-46-2404/153
loop17    7:17   0   1.7M  1 loop /snap/gnome-system-monitor/193
loop18    7:18   0   395M  1 loop /snap/mesa-2404/1165
loop19    7:19   0   2.5M  1 loop /snap/gnome-system-monitor/163
loop20    7:20   0   548K  1 loop /snap/gnome-logs/106
loop21    7:21   0 349.7M  1 loop /snap/gnome-3-38-2004/143
loop22    7:22   0   2.1M  1 loop /snap/gnome-calculator/972
sda       8:0    0   450G  0 disk
└─sda1    8:1    0   450G  0 part /
sdb       8:16   1  29.1G  0 disk
└─sdb1    8:17   1  29.1G  0 part /media/swd/BOOT
sr0      11:0    1  1024M  0 rom
swd@swd:~$
```

- 没有烧录工具，先安装烧录工具 sudo apt install dd
- 烧录镜像到 SD 卡
- sudo dd if=/dev/sdb of=xxx.img bs=4M status=progress



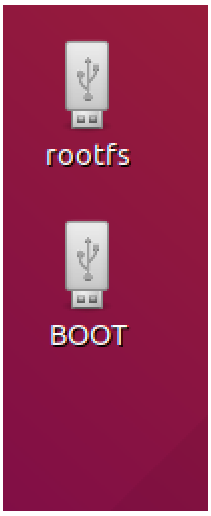
```
es  Terminal
File  Edit  View  Search  Terminal  Help
swd@swd:~$ lsblk
NAME    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
loop0     7:0    0  65.1M  1 loop /snap/gtk-common-themes/1515
loop1     7:1    0    4K   1 loop /snap/bare/5
loop2     7:2    0   2.5M  1 loop /snap/gnome-calculator/884
loop3     7:3    0   860K  1 loop /snap/gnome-logs/129
loop4     7:4    0   9.4M  1 loop /snap/gnome-characters/805
loop5     7:5    0  61.8M  1 loop /snap/core20/1081
loop6     7:6    0  55.4M  1 loop /snap/core18/2128
loop7     7:7    0 218.4M  1 loop /snap/gnome-3-34-1804/93
loop8     7:8    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop9     7:9    0   704K  1 loop /snap/gnome-characters/726
loop10    7:10   0  55.5M  1 loop /snap/core18/2979
loop11    7:11   0  66.9M  1 loop /snap/core24/1267
loop12    7:12   0 241.4M  1 loop /snap/gnome-3-38-2004/70
loop13    7:13   0   219M  1 loop /snap/gnome-3-34-1804/72
loop14    7:14   0  63.8M  1 loop /snap/core20/2686
loop15    7:15   0  48.1M  1 loop /snap/snapd/25935
loop16    7:16   0 606.1M  1 loop /snap/gnome-46-2404/153
loop17    7:17   0   1.7M  1 loop /snap/gnome-system-monitor/193
loop18    7:18   0   395M  1 loop /snap/mesa-2404/1165
loop19    7:19   0   2.5M  1 loop /snap/gnome-system-monitor/163
loop20    7:20   0   548K  1 loop /snap/gnome-logs/106
loop21    7:21   0 349.7M  1 loop /snap/gnome-3-38-2004/143
loop22    7:22   0   2.1M  1 loop /snap/gnome-calculator/972
sda       8:0    0   450G  0 disk
└─sda1    8:1    0   450G  0 part /
sdb       8:16   1  29.1G  0 disk
└─sdb1    8:17   1  29.1G  0 part /media/swd/BOOT
swd@swd:~$ sudo dd if=openwifi-1.5.0-shahecheng.img of=/dev/sdb bs=4M status=progress
```

➢ 烧录进行中，等待时间较长，笔者等待了 25 分钟左右。

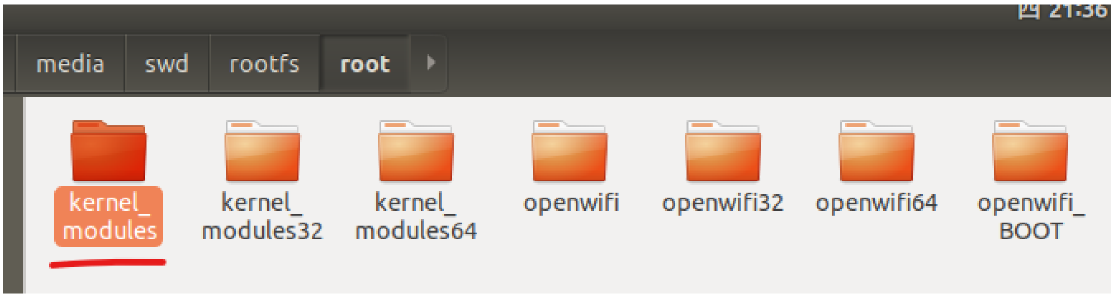


```
es  Terminal
File  Edit  View  Search  Terminal  Help
swd@swd:~$ lsblk
NAME    MAJ:MIN RM   SIZE RO TYPE MOUNTPOINT
loop0     7:0    0  65.1M  1 loop /snap/gtk-common-themes/1515
loop1     7:1    0    4K   1 loop /snap/bare/5
loop2     7:2    0   2.5M  1 loop /snap/gnome-calculator/884
loop3     7:3    0   860K  1 loop /snap/gnome-logs/129
loop4     7:4    0   9.4M  1 loop /snap/gnome-characters/805
loop5     7:5    0  61.8M  1 loop /snap/core20/1081
loop6     7:6    0  55.4M  1 loop /snap/core18/2128
loop7     7:7    0 218.4M  1 loop /snap/gnome-3-34-1804/93
loop8     7:8    0  91.7M  1 loop /snap/gtk-common-themes/1535
loop9     7:9    0   704K  1 loop /snap/gnome-characters/726
loop10    7:10   0  55.5M  1 loop /snap/core18/2979
loop11    7:11   0  66.9M  1 loop /snap/core24/1267
loop12    7:12   0 241.4M  1 loop /snap/gnome-3-38-2004/70
loop13    7:13   0   219M  1 loop /snap/gnome-3-34-1804/72
loop14    7:14   0  63.8M  1 loop /snap/core20/2686
loop15    7:15   0  48.1M  1 loop /snap/snapd/25935
loop16    7:16   0 606.1M  1 loop /snap/gnome-46-2404/153
loop17    7:17   0   1.7M  1 loop /snap/gnome-system-monitor/193
loop18    7:18   0   395M  1 loop /snap/mesa-2404/1165
loop19    7:19   0   2.5M  1 loop /snap/gnome-system-monitor/163
loop20    7:20   0   548K  1 loop /snap/gnome-logs/106
loop21    7:21   0 349.7M  1 loop /snap/gnome-3-38-2004/143
loop22    7:22   0   2.1M  1 loop /snap/gnome-calculator/972
sda       8:0    0   450G  0 disk
└─sda1    8:1    0   450G  0 part /
sdb       8:16   1  29.1G  0 disk
└─sdb1    8:17   1  29.1G  0 part /media/swd/BOOT
swd@swd:~$ sudo dd if=openwifi-1.5.0-shahecheng.img of=/dev/sdb bs=4M status=progress
[sudo] password for swd:
15921577984 bytes (16 GB, 15 GiB) copied, 1505 s, 10.6 MB/s
3798+1 records in
3798+1 records out
15931539456 bytes (16 GB, 15 GiB) copied, 1505.94 s, 10.6 MB/s
swd@swd:~$
```

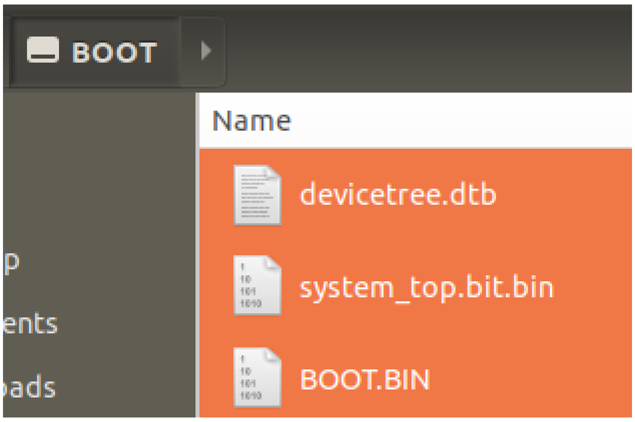
➢ 烧录完成后，在 Ubuntu 桌面会显示 SD 卡的 2 个分区，如果没有，先弹出 SD 卡，再插入。

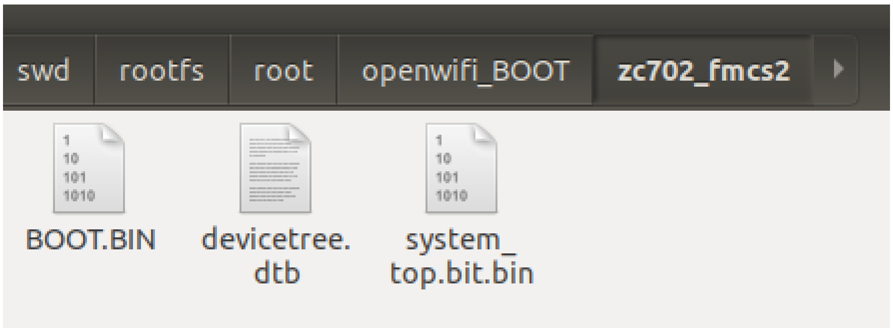


➢ 在 rootfs 分区 root 文件夹内找到 kernel_modules 文件夹，删除。

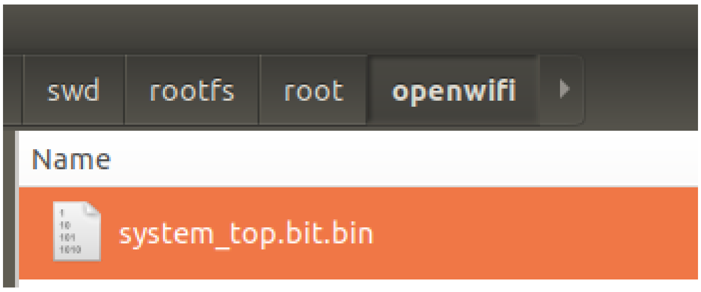


➢ 将 BOOT.BIN、system_top.bit.bin 和 devicetree.dtb 拷贝至 BOOT 分区根目录和 rootfs 分区 root/openwifi_BOOT/zc702_fmcs2 文件夹。





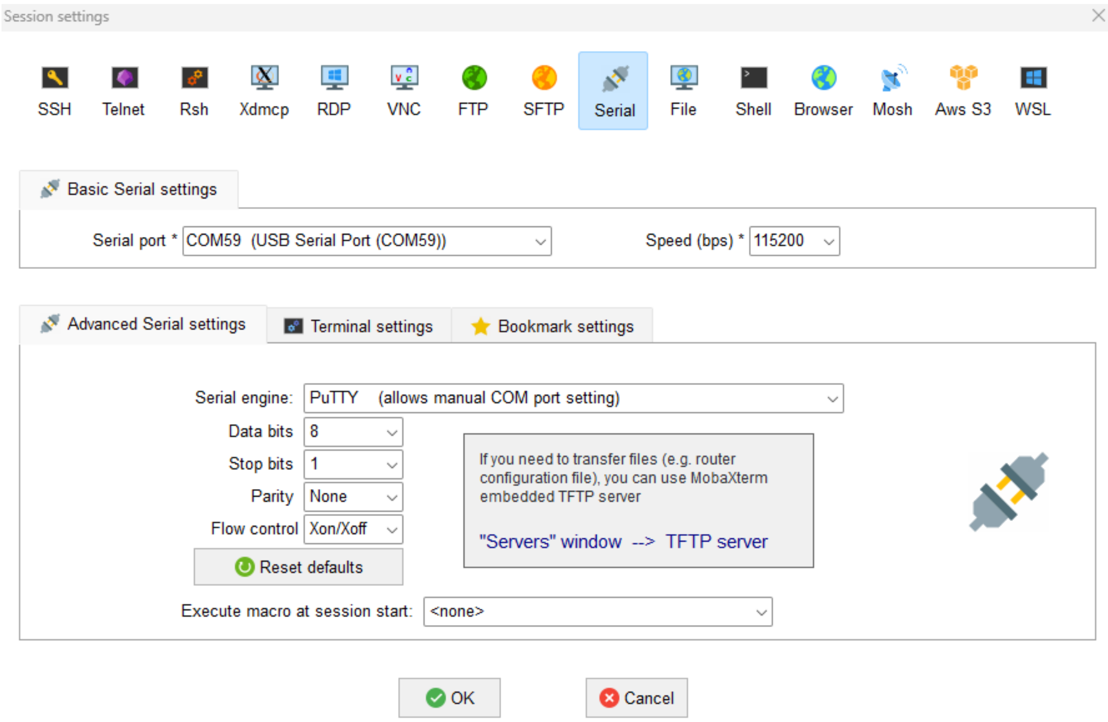
- 将 system_top.bit.bin 拷贝至 rootfs 分区/root/openwifi 文件夹内。

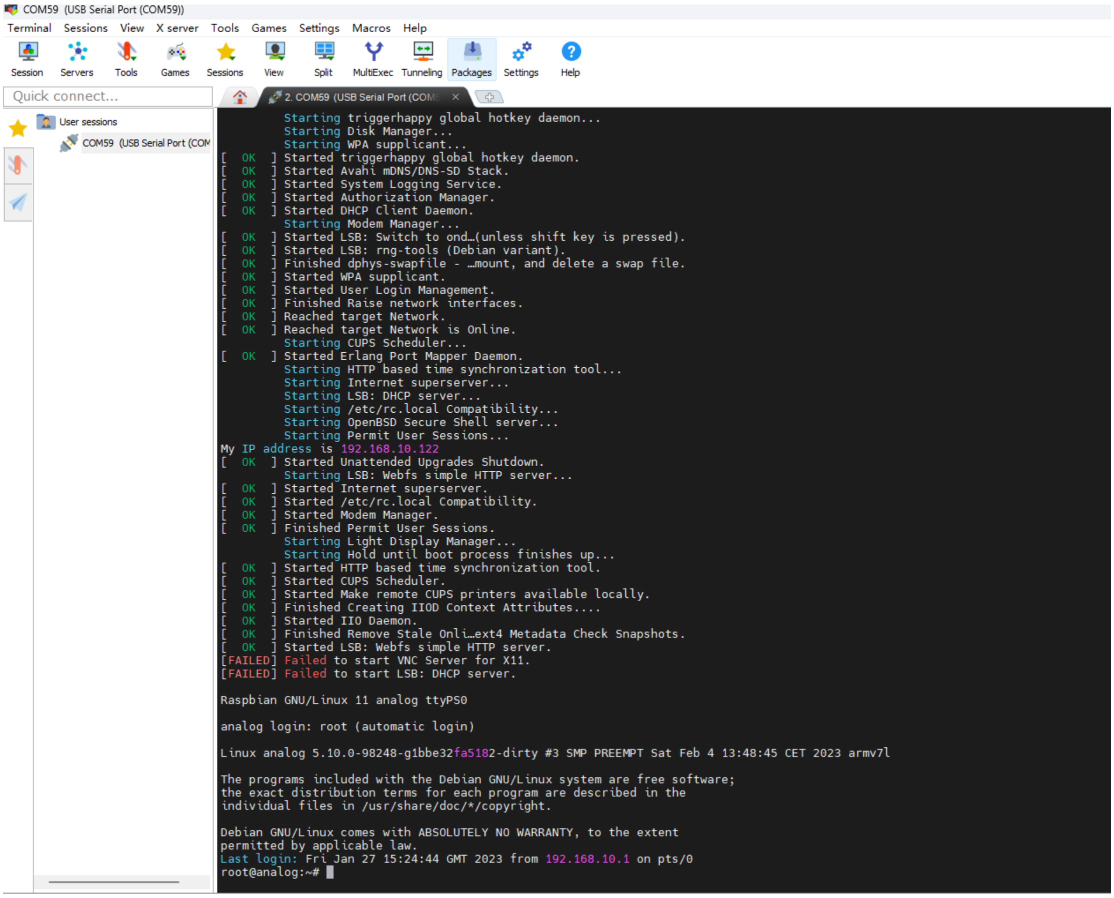


- 弹出 SD 卡，将 SD 卡插入开发板。

5 启动开发板

- 将 TypeC 线缆接入计算机 USB 口，开发板加电。
- 打开串口调试软件，连接，可以看到启动打印信息。如果此时板卡启动完成，可以复位重新启动，查看启动信息。或者回车直接执行后续指令。

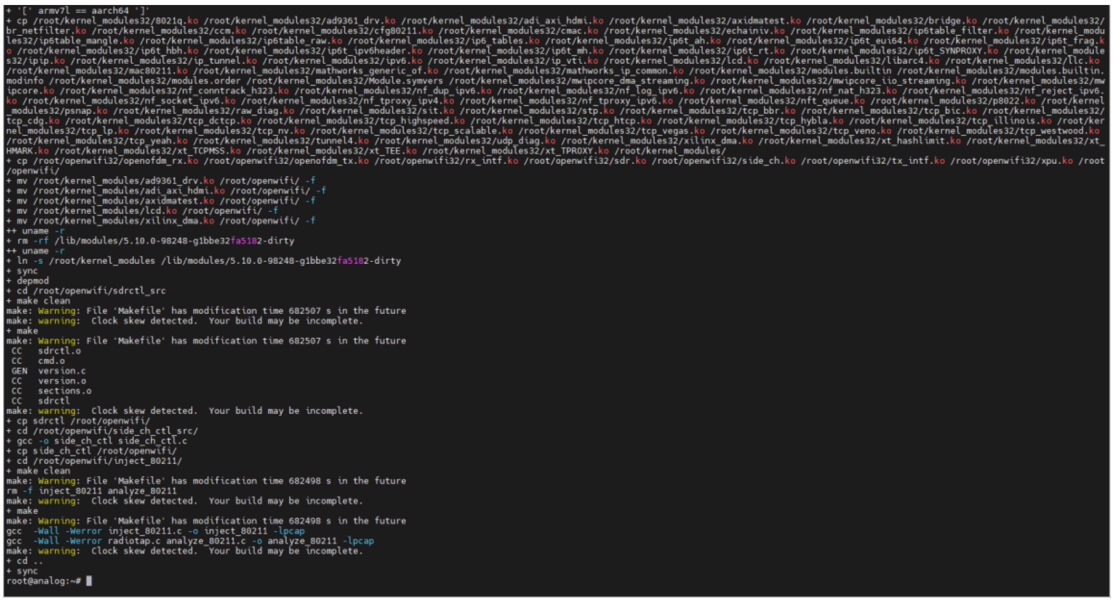




➢ 第一次启动时，输入./openwifi/setup_once.sh。

```
root@analog:~# ./openwifi/setup_once.sh
```

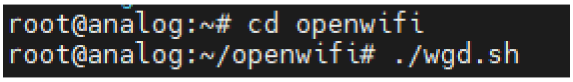
➢ 执行完后复位重启设备。



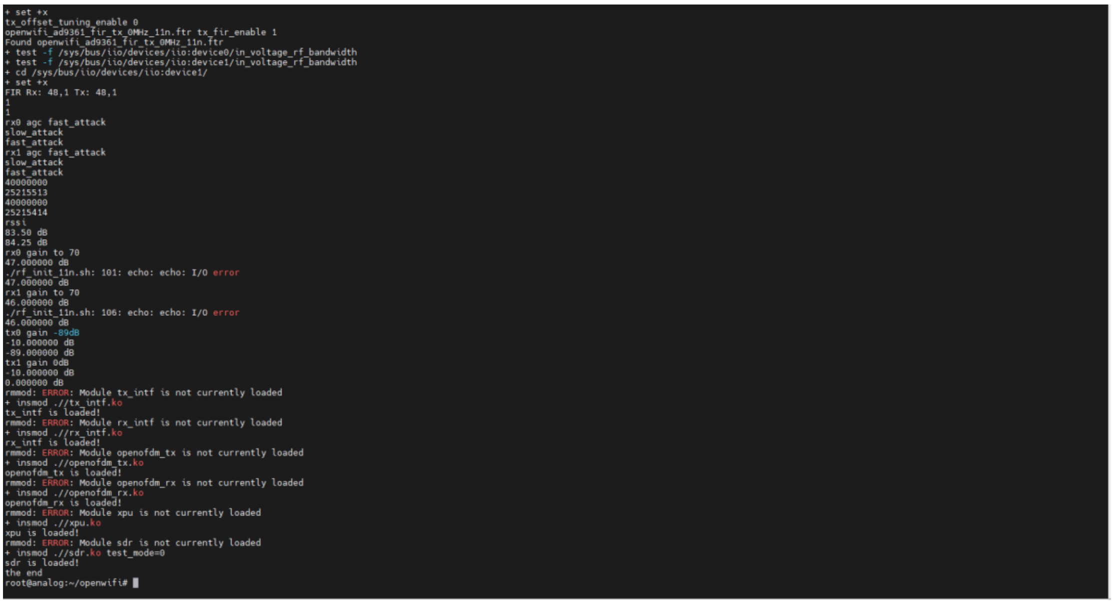
➢ 依次执行以下指令。

➢ cd openwifi

- ./wgd.sh



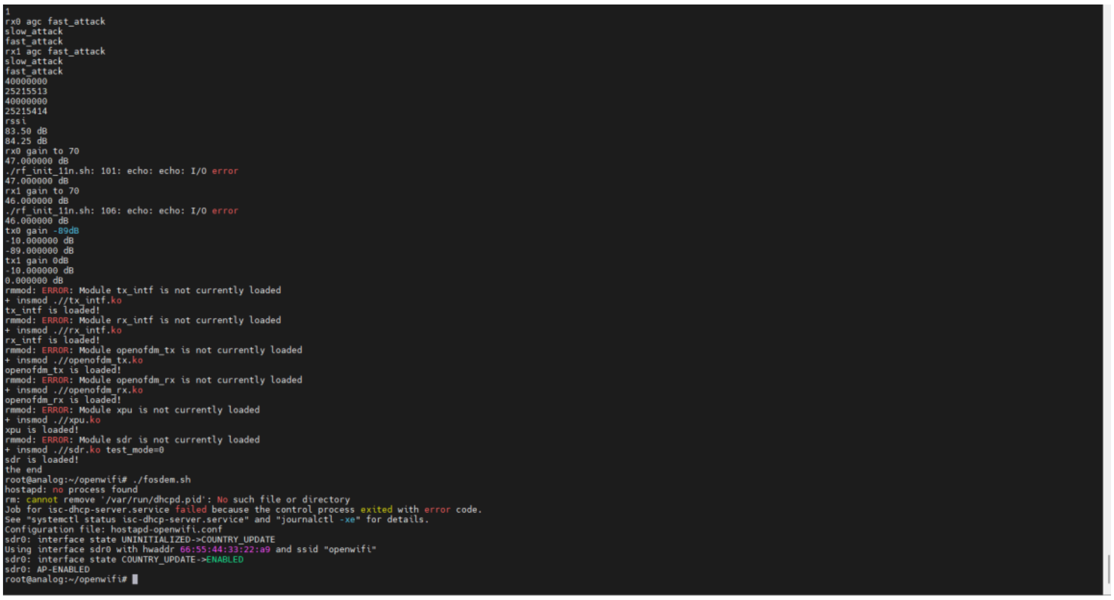
```
root@analog:~# cd openwifi
root@analog:~/openwifi# ./wgd.sh
```



```
+ set +x
tx_offset_tuning_enable 0
openwifi_ad9361_fir_tx_0MHz_11n.ftr tx_fir_enable 1
Found openwifi_ad9361_fir_tx_0MHz_11n.ftr
+ test -f /sys/bus/iio/devices/iio:device0/in_voltage_rf_bandwidth
+ test -f /sys/bus/iio/devices/iio:device1/in_voltage_rf_bandwidth
+ cd /sys/bus/iio/devices/iio:device1/
+ set +x
FIR Rx: 48,1 Tx: 48,1
1
1
rx0 agc fast_attack
slow_attack
fast_attack
rx1 agc fast_attack
slow_attack
fast_attack
40000000
25215513
40000000
25215414
rssi
83.50 dB
84.25 dB
rx0 gain to 70
47.000000 dB
./rf_init_11n.sh: 101: echo: echo: I/O error
47.000000 dB
rx1 gain to 70
46.000000 dB
./rf_init_11n.sh: 106: echo: echo: I/O error
46.000000 dB
tx0 gain -89dB
-10.000000 dB
-89.000000 dB
tx1 gain 0dB
-10.000000 dB
0.000000 dB
rmmod: ERROR: Module tx_intf is not currently loaded
+ insmod .//tx_intf.ko
tx_intf is loaded!
rmmod: ERROR: Module rx_intf is not currently loaded
+ insmod .//rx_intf.ko
rx_intf is loaded!
rmmod: ERROR: Module openofdm_tx is not currently loaded
+ insmod .//openofdm_tx.ko
openofdm_tx is loaded!
rmmod: ERROR: Module openofdm_rx is not currently loaded
+ insmod .//openofdm_rx.ko
openofdm_rx is loaded!
rmmod: ERROR: Module xpu is not currently loaded
+ insmod .//xpu.ko
xpu is loaded!
rmmod: ERROR: Module sdr is not currently loaded
+ insmod .//sdr.ko test_mode=0
sdr is loaded!
the end
root@analog:~/openwifi#
```

- ./fosdem-11ag.sh

```
root@analog:~/openwifi# ./fosdem-11ag.sh
```



```
1
rx0 agc fast_attack
slow_attack
fast_attack
rx1 agc fast_attack
slow_attack
fast_attack
40000000
25215513
40000000
25215414
rssi
83.50 dB
84.25 dB
rx0 gain to 70
47.000000 dB
./rf_init_11n.sh: 101: echo: echo: I/O error
47.000000 dB
rx1 gain to 70
46.000000 dB
./rf_init_11n.sh: 106: echo: echo: I/O error
46.000000 dB
tx0 gain -89dB
-10.000000 dB
-89.000000 dB
tx1 gain 0dB
-10.000000 dB
0.000000 dB
rmmod: ERROR: Module tx_intf is not currently loaded
+ insmod .//tx_intf.ko
tx_intf is loaded!
rmmod: ERROR: Module rx_intf is not currently loaded
+ insmod .//rx_intf.ko
rx_intf is loaded!
rmmod: ERROR: Module openofdm_tx is not currently loaded
+ insmod .//openofdm_tx.ko
openofdm_tx is loaded!
rmmod: ERROR: Module openofdm_rx is not currently loaded
+ insmod .//openofdm_rx.ko
openofdm_rx is loaded!
rmmod: ERROR: Module xpu is not currently loaded
+ insmod .//xpu.ko
xpu is loaded!
rmmod: ERROR: Module sdr is not currently loaded
+ insmod .//sdr.ko test_mode=0
sdr is loaded!
the end
root@analog:~/openwifi# ./fosdem.sh
hostapd: no process found
rm: cannot remove '/var/run/dhcpd.pid': No such file or directory
Job for isc-dhcp-server.service failed because the control process exited with error code.
See "systemctl status isc-dhcp-server.service" and "journalctl -xe" for details.
Configuration file: hostapd-openwifi.conf
sdr0: interface state UNINITIALIZED->COUNTRY_UPDATE
Using interface sdr0 with hwaddr 66:55:44:33:22:a9 and ssid "openwifi"
sdr0: interface state COUNTRY_UPDATE->ENABLED
sdr0: AP-ENABLED
root@analog:~/openwifi#
```

- 最后提示 AP 开启，启动完成。

6 openwifi 连接测试

- PC 连接 openwifi，但此时还不能联网。

7 openwifi 联网测试

- 串口输入 ifconfig 查看开发板接口，eth0 为板卡有线接口，sdr0 为板卡无线接口。

```
root@analog:~/openwifi# ifconfig
eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.10.122  netmask 255.255.255.0  broadcast 192.168.10.255
        inet6 fe80::2658:b703:c1cb:9fe3  prefixlen 64  scopeid 0x20<link>
        ether 00:0a:35:00:01:22  txqueuelen 1000  (Ethernet)
        RX packets 6903  bytes 5078075 (4.8 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 8676  bytes 1763921 (1.6 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
        device interrupt 36  base 0xb000

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 428  bytes 27806 (27.1 KiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 428  bytes 27806 (27.1 KiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

sdr0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 192.168.13.1  netmask 255.255.255.0  broadcast 192.168.13.255
        inet6 fe80::6455:44ff:fe33:226d  prefixlen 64  scopeid 0x20<link>
        ether 66:55:44:33:22:6d  txqueuelen 1000  (Ethernet)
        RX packets 8259  bytes 1759335 (1.6 MiB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 6859  bytes 5347434 (5.0 MiB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0
```

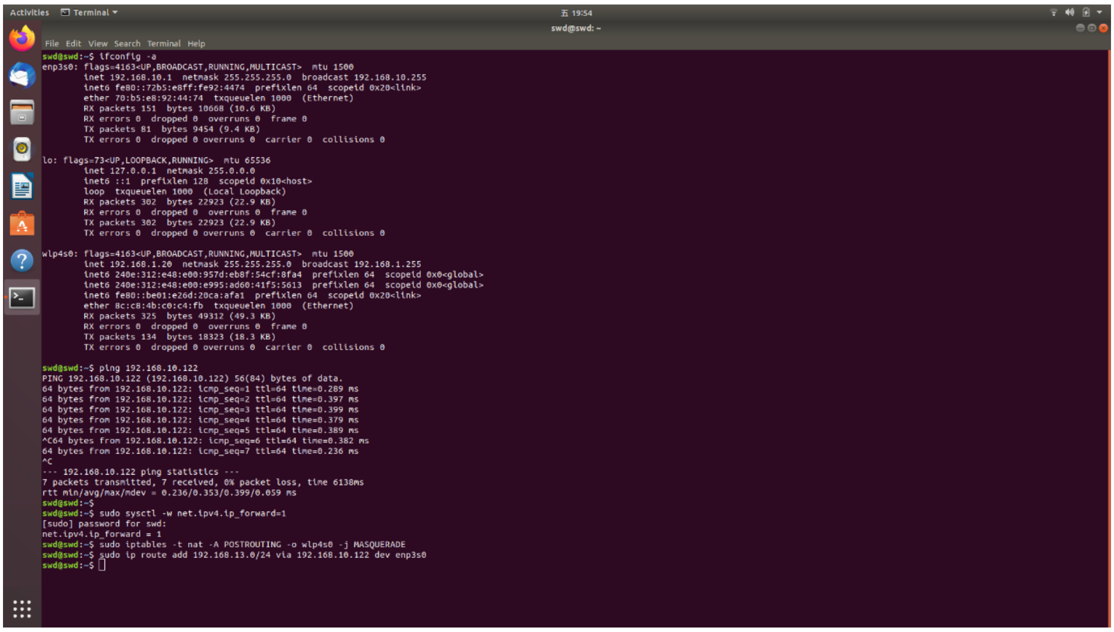
- 将开发板网口通过网线与 Ubuntu PC 连接，Ubuntu PC 有线网卡 IP 地址改为 192.168.10.1。无线网卡可联网。
- 打开终端，按照下列指令配置。

- To give the Wi-Fi client internet access, configure routing/NAT **on the PC**:

```
sudo sysctl -w net.ipv4.ip_forward=1
sudo iptables -t nat -A POSTROUTING -o NICY -j MASQUERADE
sudo ip route add 192.168.13.0/24 via 192.168.10.122 dev ethX
```

ethX is the PC NIC name connecting the board ethernet. **NICY** is the PC NIC name connecting internet (WiFi or another ethernet).

➢ enps30 为笔者 Ubuntu PC 有线网卡，与板卡相连。wlp4s0 为无线网卡，可上网。



➢ 配置完成后 openwifi 可正常访问互联网。